Week 12 Notes & Quiz

Notes for all three Week 12 lectures, made only from the 97 slides in *WEEK 12.pdf*. Each yellow tag such as S43 is the slide (PDF page) the point comes from. The 40-question practice questions and answer key are at the end.

STOP	LECTURE	SLIDES
L58	**Sustainable Supply Chain**	S1–S33
L59	**Emergency Supply Chain**	S34–S68
L60	**Assets Management in Supply Chain**	S69–S97
+	**Cheat sheet · Slide quirks · Quiz (40 Qs)**	–

LECTURE 58 S1–S33

Sustainable Supply Chain

- **Supply chain sustainability**: considering the environmental and human impact of a product's journey from raw materials to delivery; minimize harm from energy, water and waste while helping people and communities, on top of revenue and profit. S3

> SUSTAINABLE DEVELOPMENT · BRUNDTLAND COMMISSION (UN) S5
>
> *"development that meets the needs of the present without compromising the ability of future generations to meet their own needs."*
>
> The **2005 UN World Summit** named **economic, environmental and social** sustainability as the "three pillars"; all three must be reconciled.

- Growth of Brazil, China, India raises living standards but pressures resources. Three drivers of focus: **reduce risk and improve financial performance**, **attract customers who value sustainability**, **make the world more sustainable**. S6
- **Unilever**: drip irrigation for Brazilian tomato growers; almost half of sales from emerging economies; buys about **10% of the world's tea** and **30% of all spinach**. S7
- **Wal-Mart**: started as a defensive move; efficient bulbs, skylights, less packaging; with **Costco** redesigned the one-gallon milk jug, saving **10–20 cents a gallon**. S8

Triple Bottom Line (TBL)

- Coined by **John Elkington**; a yardstick for how well a chain builds sustainability while staying profitable. Aims: transparency, economic opportunity, local competitiveness, less environmental degradation. S9
- Metrics: **social** (health, gender equality, labour rights), **environmental** (certified suppliers, fossil fuel, waste, electricity), **economic** (financial flow, ROI, investments). S10
- Benefits: lower carbon footprint and energy cost, better brand and productivity, future growth and financial security. S11
- Five business areas that benefit: supply chain operations, branding (MIT Sloan: consumers may pay **2–10% more** for transparent products), compliance (UN 2030 SDGs), investor relations (nearly half of investors interested in sustainable funds, Gallup), corporate culture (millennials). S12–S13

10 key sustainability metrics

Carbon footprint · energy consumption · water usage · waste generation · supplier diversity · ethical sourcing · supplier compliance (Fair Trade, **FSC**) · product lifecycle analysis · transportation efficiency (MPG, kg CO_2 per ton-km) · social impact. S14–S15

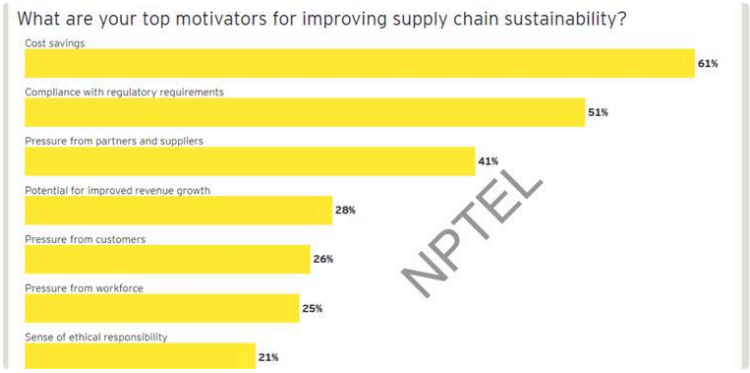
Drivers that add to sustainability

Facilities	Walmart's energy-efficient stores; waste management turned into profit. S17
Inventory	Track landfill inventory; Walmart cut phosphates in detergents and redesigned packaging. S17
Transportation	Walmart cut delivery fuel use by **65%**; IKEA designs for high volume and weight density. S17
Sourcing	Starbucks' **C.A.F.E.** program; Mattel's lead-paint toy recall shows the reputational risk. S18
Information	No standard reporting; Walmart's packaging scorecard. S18
Pricing	Time-of-day pricing; tax credits (e.g. **Nissan Leaf**); customers' willingness to pay drives long-term change. S18

7 ways to improve sustainability

1. Find pain points: calculate the **product carbon footprint (PCF)** first. S20
2. Optimize transportation: it was about **29–30% of US GHG emissions in 2019** (largest sector; electricity 25%, industry 23%); fill trucks (3D load software), EVs, route optimization. S20–S21
3. Collaborate with sustainable suppliers: standards **GRI 408 and 409**, SASB human capital, **ISO 14001**; third-party verified ESG reports and PCFs. S22
4. Minimize inventory: warehouses are about **13%** of SC-related GHG emissions. S23
5. Reuse waste or by-products (bottle deposit schemes). S23
6. Eco-friendly packaging: only **1% of plastics are biodegradable** (EU); the **Higg index** gives material-level GHG data. S24
7. Nudge regulators and consumers: Starbucks discounts for own cups, aiming to cut single-use plastic **50%**. S25

- Challenges: **cost is the primary barrier** (worse for small firms), finding sustainable components, inherited supply chains from acquisitions, organization structure, lack of customer interest. S26
- Trends: **circular economy** (eliminate waste, design for reuse) and **data-driven supply chains** (cloud, IoT). S27



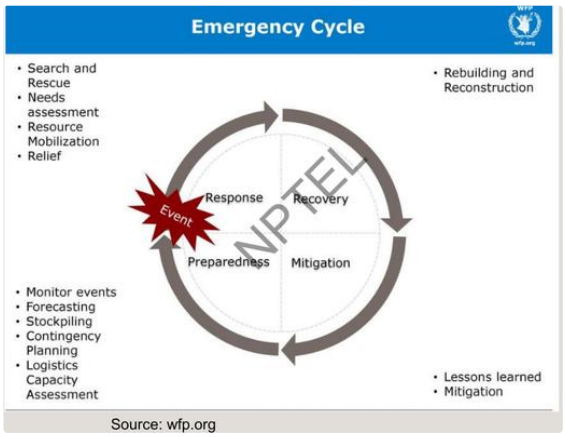
Original slide S30 (EY survey). Top motivators: **cost savings 61%**, compliance 51%, pressure from partners and suppliers 41%, revenue growth 28%, customers 26%, workforce 25%, ethical responsibility 21%.

- EY case: **8 in 10** executives are increasing sustainability efforts, but only about **4 in 10** have seen better visibility. 55% expect better operational risk management in 1–3 years (20% already); 31% already have better efficiency (51% expect it); 54% expect higher share price in 3–5 years. Five actions include aligning with commitments, visibility and traceability, wider ROI measurement, tax incentives and grants. S28–S29

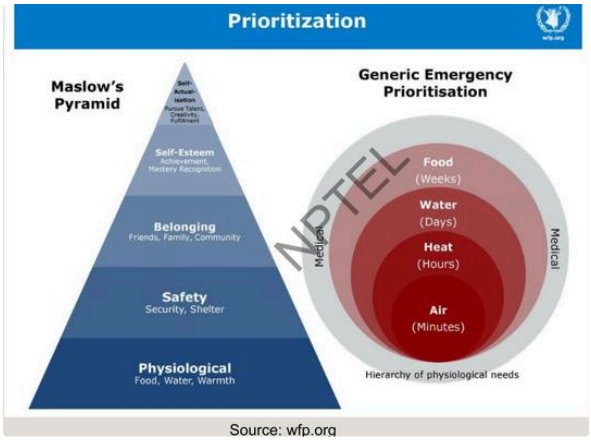
LECTURE 59 S34–S68

Emergency Supply Chain

- The Joint Logistics Enterprise defines a **complex emergency** as international conflict, humanitarian and domestic disaster relief scenarios combining warfare, food insecurity, epidemics and social conflict. S36
- **Emergency supply chain**: a specialized framework for urgent, unforeseen situations (disasters, health emergencies, geopolitical crises): rapid mobilization, coordination, deployment of goods. Two domains: **military expeditions** and **non-governmental disaster relief**. S37



Original slide S38 (WFP). Emergency cycle: **Preparedness** (forecasting, stockpiling, contingency plans) → event → **Response** (search and rescue, needs assessment, relief) → **Recovery** (rebuilding) → **Mitigation** (lessons learned).



Original slide S40 (WFP). Prioritization follows Maslow's pyramid. Survival time without: **air (minutes), heat (hours), water (days), food (weeks)**, with medical needs around all.

Why a separate framework

- A **disruption** hits an existing chain whose identity (stakeholders, products, facilities) stays the same. An **emergency** supply chain is **formed in response** to the emergency, often without prior planning. S41–S42
- Emergency chains prioritize **effectiveness over efficiency**: an "**at all cost**" approach; fly supplies instead of sealift; more less-than-truckload shipments. Conventional chains maximize service at minimum cost. S43–S44

Uniqueness of emergency logistics 4 items

1. Diverse, immediate services needed (water, food, medicine, shelter); timing drives survival rates. S45
2. Real-time demand information is hard to get (affected people can't report). S46
3. **Service efficiency over profit**, e.g. costly air transport. S46
4. Collaboration among government, public bodies, firms, individuals. S46

- Solutions: governments set up an emergency SCM system in normal times; invest in labour, R&D, IT and software; strengthen communication to cut information distortion and the **bullwhip effect**; collaborative mechanisms like electronic procurement hubs. S47–S48

Humanitarian supply chain management

- Buying and delivering requested supplies and services where and when needed, with **best value for money**, after an emergency. Technical side: planning, information, warehousing, **pre-positioning**, transport, distribution. Strategic side: decisions, coordination, partnerships. S49
- Examples: **2015 Nepal earthquake** (helicopters), **Hurricane Maria, Puerto Rico 2017** (power grid restoration), **2010 Haiti earthquake** (pre-positioned supplies), **Za'atari refugee camp**, Jordan, sub-Saharan healthcare. S50–S57
- Challenges: unpredictable demand (**Ebola** in West Africa), limited access (**Rohingya** camps in Bangladesh), domestic barriers (**Syrian** crisis border crossings); also lack of trust, many organizations, inefficient collaboration, fraud, centralization, cost inefficiencies. S52–S55
- 60–80% of the cost of an emergency response is within the supply chain; 30 humanitarian organizations spend about **$6 billion** a year on supply chains. S57
- General steps: **needs assessment → planning → sourcing → procurement → transportation → warehousing → distribution**, plus monitoring. S58–S59
- WFP elements: planning, sourcing, logistics, efficiency, effectiveness, with information along the chain. Cargo: relief items (medical, shelter, WASH, dangerous goods, equipment, food) and food commodities (cereals, oils, salt, sugar, pulses, fortified foods, RTUF). S60–S61

Case: Australia's risky food supply chain

- Floods in Queensland and New South Wales emptied shelves; supermarkets introduced buying limits; the **IPCC** warns extreme weather will become more frequent. S63
- Governments rely mainly on the **food industry** for resilience and on **charities** for emergency relief; Brisbane Markets closed, Pacific Highway blocked. Melbourne research identifies features of a resilient food system (closed-loop agriculture, circular economy, decentralized and local chains). S64–S66

LECTURE 60 S69–S97

Assets Management in Supply Chain

- A **supply chain asset** is any tangible or intangible resource that contributes to the chain's functioning: facilities, warehouses, vehicles, inventory, information systems, IP, supplier relationships, brand. S71
- Supply chain assets are those directly attributable to supply, manufacture and distribution; they **exclude cash, cash equivalents and liquid investments**. S72
- **Asset management** (Corporate Finance Institute): developing, operating, maintaining and selling assets cost-effectively; includes scheduled maintenance, work orders, historical data, scheduling and institutional knowledge. Reactive-only management stalls productivity. S74–S75

INVENTORY MANAGEMENT	ASSET MANAGEMENT S77
Tracks stock entering and leaving	Manages items the firm keeps for their whole life cycle, procurement to disposal
Replenishes goods to sell	Inspection, cleaning, repair or replacement
Returns only if customers are dissatisfied	Users borrow and return items to the asset pool
Minimal detail per item	Decides which assets add value and which are liabilities

- Types: raw materials, WIP, finished goods, packaging, tools and equipment, vehicles, warehousing, IT assets, intellectual property, human resources. S78–S79

Asset management KPIs

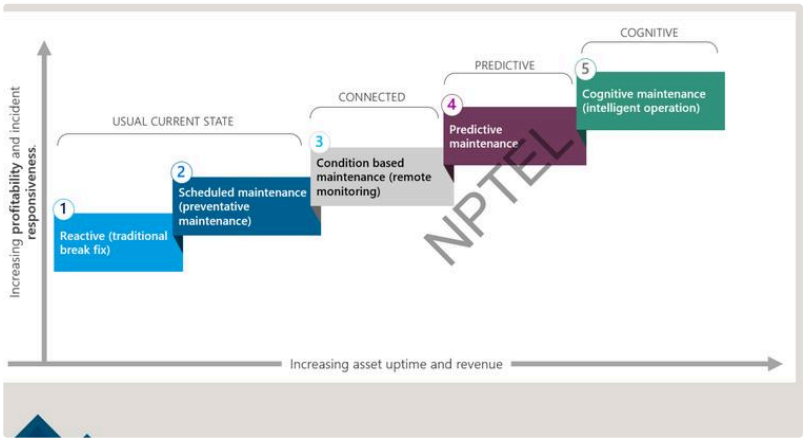
MTBF	Mean time between failures = (time in use − unplanned downtime) ÷ number of breakdowns. S80
MTTR	Mean time to repair = total maintenance time ÷ number of repairs; compares technicians. S80
OEE	Overall Equipment Effectiveness = Availability × Performance × Quality; availability = true ÷ planned operating time; performance = actual ÷ planned cycle time; quality = valid ÷ total pieces. S80
Others	Asset utilization rate, ROA, maintenance cost per asset, downtime, availability, **life cycle cost**, performance ratio, work order completion rate, **Asset Condition Index (ACI)**. S81

Asset tracking

- End-to-end tracking gives cost-effective monitoring, real-time visibility, traceability (cold chains), sustainability optimization, control. Not adopting it risks inefficiency, disruptions, asset loss and losing competitiveness. S82–S83
- Technologies: **RFID, GPS, barcodes, IoT sensors, NFC, BLE** (indoor), **RTLS** (real-time location systems), cloud software, blockchain, AI/ML, mobile apps, drones, edge computing, AR, voice recognition, thermal imaging, ultrasound (indoor where GPS fails), smart packaging, **telematics** (fleet), wearables. S84–S87
- Best practices: comprehensive inventory of assets, standardized tracking, centralized database, regular audits, maintenance protocols, real-time monitoring, preventive maintenance, lifecycle management, vendor management, training, analytics, continuous improvement. S88–S89
- Future trends (11): IoT, predictive maintenance, blockchain, AI, autonomous vehicles and drones, AR for maintenance, **digital twins**, sustainable asset management, cloud systems, robotics, multi-tier visibility. S90–S91

Case: EAM in Microsoft Dynamics 365

- Asset Management announced in the **October 2019** release; Microsoft ERP solutions: D365 Finance, D365 Supply Chain Management, D365 Business Central. S92–S93



Original slide S94. Five maintenance models, rising in uptime and value: **reactive** (break-fix) → **scheduled** (preventive) → **condition-based** (remote monitoring) → **predictive** → **cognitive** (intelligent operation).

BEFORE THE EXAM ALL LECTURES

Cheat sheet

Brundtland	Meet present needs without compromising future generations S5
Three pillars (2005)	Economic, environmental, social S5
TBL	John Elkington; social, environmental, economic measures S9
Numbers	Unilever 10% tea, 30% spinach · milk jug 10–20¢/gal · Walmart fuel −65% · transport ~29–30% of US GHG · warehouses 13% · 1% plastics biodegradable · cost savings 61% top motivator
Emergency SC	Effectiveness over efficiency, "at all cost"; formed in response, not pre-existing S41–S44
Emergency cycle	Preparedness → response → recovery → mitigation S38
Priorities	Air (minutes) → heat (hours) → water (days) → food (weeks) S40
Humanitarian SC cost	60–80% of response cost; ~$6 bn/yr across 30 organizations S57
Asset KPIs	MTBF, MTTR, OEE = A × P × Q S80
5 maintenance models	Reactive → scheduled → condition-based → predictive → cognitive S93–S94

WORKED EXAMPLE (MY OWN NUMBERS, USING THE SLIDE FORMULAS)

Availability 90%, performance 95%, quality 98% → OEE = 0.90 × 0.95 × 0.98 ≈ **83.8%**. A machine used 1,000 h with 40 h unplanned downtime and 4 breakdowns → MTBF = 960 ÷ 4 = **240 h**; 4 repairs taking 20 h → MTTR = **5 h**.

READ ONCE SOURCE CHECK

Slide quirks to watch

- **Transport's share of US GHG (2019):** the text says "about 30%" while the pie chart shows 29%. S20–S21
- The OEE slide defines performance as "actual cycle time / planned cycle time", which is inverted from the usual ideal ÷ actual cycle time. Use the slide's wording if asked. S80
- "Uniliver" in the slide title is Unilever. S7
- The asset management benefits slide repeats "reduce their carbon footprint" under "enhanced control". S82
- The Lecture 58 references list (S32) is the same as Week 11's digitalization references and includes an unrelated physics paper.
- Video or link-only slides: S4, S39, S73.

SELF-TEST 40 QUESTIONS

Week 12 practice questions

Answer these first, then check the key on the next pages. Option letters match the online quiz.

Q1 LECTURE 58

The Brundtland Commission defined sustainable development as development that:

A meets present needs without compromising future generations' ability to meet theirs
B focuses only on the environment
C stops all growth
D maximizes present profits

Q2 LECTURE 58

The 2005 UN World Summit identified which three pillars of sustainable development?

A Plan, do, check
B People, process, technology
C Economic, environmental, social
D Cost, quality, time

Q3 LECTURE 58

The Triple Bottom Line concept was coined by:

A John Elkington
B Michael Porter
C Peter Drucker
D Keith Oliver

Q4 LECTURE 58

Unilever buys roughly what share of the world's tea crop?

A 5%
B 10%
C 30%
D 50%

Q5 LECTURE 58

Wal-Mart and Costco's redesigned one-gallon milk jug saved about:

A 50 cents a gallon
B 10–20 cents a gallon
C $1 a gallon
D 1–2 cents a gallon

Q6 LECTURE 58

MIT Sloan researchers found consumers may pay how much more for products with supply chain transparency?

A 0.5–1%
B 15–20%
C 2–10%
D 25–30%

Q7 LECTURE 58

Walmart decreased fuel usage for product delivery by about:

A 65%
B 35%
C 15%
D 90%

Q8 LECTURE 58

Mattel's toy recall due to lead paint is cited as a sustainability failure in which driver?

A Information
B Sourcing
C Pricing
D Facilities

Q9 LECTURE 58

What is the first step to improving supply chain sustainability?

A Switch to EVs
B Calculate the product carbon footprint to find pain points
C Close warehouses
D Buy carbon credits

Q10 LECTURE 58

In 2019, which sector had the largest share of US GHG emissions?

A Industry
B Transportation
C Electricity
D Agriculture

Q11 LECTURE 58

Warehouses account for about what share of supply chain-related GHG emissions?

A 30%
B 50%
C 13%
D 3%

Q12 LECTURE 58

According to the EU, what share of plastics are biodegradable?

A 1%
B 50%
C 10%
D 25%

Q13 LECTURE 58

Which standard is cited for environmental measures when vetting suppliers?

A ISO 9001
B ISO 14001
C ISO 27001
D ISO 45001

Q14 LECTURE 58

The primary barrier to implementing sustainable supply chains is:

A cost
B lack of technology
C regulation
D customer demand

Q15 LECTURE 58

In the EY survey, the top motivator for improving supply chain sustainability was:

A Ethical responsibility (21%)
B Customer pressure (26%)
C Cost savings (61%)
D Regulatory compliance (51%)

Q16 LECTURE 59

Emergency supply chains prioritize:

A efficiency over effectiveness
B lowest cost
C effectiveness over efficiency
D maximum inventory

Q17 LECTURE 59

The key difference between a disruption and an emergency supply chain is that:

A disruptions create new stakeholders
B disruptions never cost money
C an emergency supply chain is formed in response to the emergency rather than existing beforehand
D emergencies only affect military

Q18 LECTURE 59

Emergency supply chains are composed of which two domains?

A Air and sea
B Public and private banks
C Military expeditions and non-governmental disaster relief
D Retail and wholesale

Q19 LECTURE 59

In the WFP emergency cycle, stockpiling, forecasting and contingency planning belong to:

A Mitigation
B Response
C Recovery
D Preparedness

Q20 LECTURE 59

In WFP's emergency prioritization, survival without which need is measured in minutes?

A Food
B Water
C Heat
D Air

Q21 LECTURE 59

Which is a unique feature of emergency logistics compared with business logistics?

A Accurate real-time demand information is hard to obtain
B Only one organization is involved
C Profit is the main goal
D Demand information is actively provided by customers

Q22 LECTURE 59

Better communication among government levels in emergencies helps reduce:

A tariffs
B inventory turns
C the bullwhip effect
D economies of scale

Q23 LECTURE 59

Humanitarian supply chain management aims to deliver supplies where and when needed while ensuring:

- **A** best value for money
- **B** single sourcing
- **C** minimum speed
- **D** maximum profit

Q24 LECTURE 59

What share of the cost of an emergency response lies within the supply chain?

- **A** 60–80%
- **B** 95%
- **C** 10–20%
- **D** 30–40%

Q25 LECTURE 59

The Ebola outbreak in West Africa illustrates which humanitarian SC challenge?

- **A** Unpredictable demand
- **B** Domestic barriers
- **C** Fraud
- **D** Limited access

Q26 LECTURE 59

What is the first general step of the humanitarian supply chain?

- **A** Warehousing
- **B** Needs assessment
- **C** Distribution
- **D** Procurement

Q27 LECTURE 59

Which disaster response used helicopters and unconventional methods to deliver aid, per the slides?

- **A** 2010 Haiti earthquake
- **B** Hurricane Maria 2017
- **C** Za'atari camp
- **D** 2015 Nepal earthquake

Q28 LECTURE 59

In Australia's food supply chain case, governments mainly rely on ____ for resilience and ____ for emergency relief.

- **A** the food industry; charities
- **B** the military; the UN
- **C** farmers; banks
- **D** imports; exports

Q29 LECTURE 60

Which is NOT included in supply chain assets per the slides?

- **A** Information systems
- **B** Cash and cash equivalents
- **C** Warehouses
- **D** Transportation vehicles

Q30 LECTURE 60

Asset management differs from inventory management because it:

- **A** only handles returns
- **B** tracks stock to sell
- **C** records minimal detail
- **D** manages items the firm keeps through their whole life cycle

Q31 LECTURE 60

MTBF is calculated as:

- **A** availability × performance × quality
- **B** (time in use − unplanned downtime) ÷ number of breakdowns
- **C** valid pieces ÷ total pieces
- **D** total maintenance time ÷ number of repairs

Q32 LECTURE 60

Overall Equipment Effectiveness (OEE) equals:

- **A** Availability × Performance × Quality
- **B** Availability + Performance + Quality
- **C** Uptime ÷ downtime
- **D** MTBF ÷ MTTR

Q33 LECTURE 60

With availability 90%, performance 95% and quality 98%, OEE is about:

- **A** 90%
- **B** 94.3%
- **C** 83.8%
- **D** 283%

Q34 LECTURE 60

A machine was in use 1,000 hours with 40 hours of unplanned downtime and 4 breakdowns. MTBF is:

- **A** 960 h
- **B** 240 h
- **C** 10 h
- **D** 250 h

Q35 LECTURE 60

Which tracking technology suits indoor environments over short distances?

- **A** Drones
- **B** GPS
- **C** Satellite telematics
- **D** Bluetooth Low Energy (BLE)

Q36 LECTURE 60

RTLS stands for:

- **A** Real-Time Location Systems
- **B** Remote Transport Logistics Service
- **C** Route and Time Logging Software
- **D** Rapid Tracking and Labelling Standard

Q37 LECTURE 60

Systems integrating GPS, cellular and onboard diagnostics to track vehicles are called:

- **A** telematics
- **B** NFC
- **C** edge computing
- **D** smart packaging

Q38 LECTURE 60

Microsoft announced Asset Management for Dynamics 365 in which release?

- **A** October 2017
- **B** April 2021
- **C** January 2023
- **D** October 2019

Q39 LECTURE 60

Which is the most advanced of the five maintenance models?

- **A** Scheduled
- **B** Predictive
- **C** Cognitive
- **D** Reactive

ANSWERS 40 QUESTIONS

Answer key

Q	ANS	WHY · SLIDE
Q1	A	S5. S5
Q2	C	All three must be reconciled. S5
Q3	A	S9. S9
Q4	B	And about 30% of all spinach. S7
Q5	B	S8. S8
Q6	C	S12. S12
Q7	A	Through efficient loading and better fuel efficiency. S17
Q8	B	Starbucks' C.A.F.E. is the positive sourcing example. S18
Q9	B	S20. S20
Q10	B	Transportation 29%, electricity 25%, industry 23%, commercial and residential 13%, agriculture 10%. S21
Q11	C	S23. S23
Q12	A	The Higg index helps choose lower-GHG packaging materials. S24
Q13	B	Also GRI 408 and 409 and SASB human capital. S22
Q14	A	Smaller companies struggle more with upfront costs. S26
Q15	C	S30. S30
Q16	C	An 'at all cost' approach. S43
Q17	C	In a disruption, the chain's identity stays the same. S41
Q18	C	S37. S37
Q19	D	Response: search and rescue, needs assessment, relief. S38
Q20	D	Air (minutes), heat (hours), water (days), food (weeks). S40
Q21	A	Affected people often cannot report needs. S46
Q22	C	By reducing information distortion. S48
Q23	A	S49. S49
Q24	A	30 organizations spend about $6 billion a year on supply chains. S57
Q25	A	Rohingya → limited access; Syria → domestic barriers. S52
Q26	B	Needs assessment → planning → sourcing → procurement → transportation → warehousing → distribution. S58
Q27	D	S50. S50
Q28	A	S64. S64
Q29	B	Cash, cash equivalents and liquid investments are excluded. S72
Q30	D	Procurement to disposal; users borrow and return items. S77
Q31	B	MTTR = total maintenance time ÷ number of repairs. S80
Q32	A	S80. S80
Q33	C	$0.90 \times 0.95 \times 0.98 \approx 0.838$. S80
Q34	B	$(1{,}000 - 40) \div 4 = 240$ h. S80
Q35	D	Ultrasound is another indoor option where GPS fails. S85
Q36	A	Combines RFID, GPS and Wi-Fi within a confined area. S85
Q37	A	S87. S87
Q38	D	S92. S92
Q39	C	Reactive → scheduled → condition-based → predictive → cognitive. S93–S94
Q40	D	Scheduled maintenance is preventive; cognitive is intelligent operation. S94

Made from *WEEK 12.pdf* (97 slides, NPTEL Logistics & Supply Chain Management, Prof. Vikas Thakur, IIT Kharagpur). Notes use only the slides, except boxes marked "my own".